# Chi-Sheng (Michael) Chen (陳麒升)

**ML Researcher & Engineer | Clinical Biosignal ML | LLM Agents & Multi-Agent Systems | Foundation Models**
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Nationality: Taiwan | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen is a machine-learning researcher and engineer working across clinical biosignal ML and LLM-agent systems. At Harvard/BIDMC he builds multimodal EMS trauma-triage pipelines (ASR → LLM-agent event extraction → structured prediction) over 4M+ patient records; his EEG-based depression-treatment model is in routine outpatient use at Taipei Veterans General Hospital. He has shipped LLM-agent systems (a US-stock research agent suite, a robotics workcell optimizer), cut LLM token usage by 82% for a quant-trading firm, and placed 3rd of 1,300+ teams at Berkeley RDI AgentX-AgentBeats 2026. Prior: Co-Founder/CTO of Neuro Industry (GCP MLOps on 10K+ EEG samples) and Digital IC R&D Engineer at MediaTek. 49 publications, 491 citations (h-index 13); reviewer for NeurIPS, ICML (Gold Reviewer) and 40 journals.

## Research & Professional Experience

**Harvard Medical School & Beth Israel Deaconess Medical Center** — MA, USA
*Research Affiliate (primary appointment), Advisor: Prof. Dr. Gabriel Brat* — *Nov 2024 – Present*
- Building a real-time EMS trauma-triage pipeline (ASR → LLM-agent event extraction → structured prediction) over **4M+ patient records** using multimodal AI.
- Co-developed webcam-based open-surgical skill assessment (hand keypoint detection; CLINICCAI 2026 poster) and a capture-clock methodology for browser webcam gaze (DEMI 2026 spotlight talk).

**WhaleForce (subsidiary of VICI Holdings)** — Taipei, Taiwan
*Fractional Token Efficiency Research Lead (Advisor)* — *Aug 2026 – Present*
- Advising on LLM token-efficiency research and multi-agent architecture for AI-driven quantitative trading; token-efficiency work **cut LLM token usage by 82%**.

**Omnis Labs** — Remote
*Co-Founder* — *Sep 2024 – May 2026*
- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

**Neuro Industry, Inc.** — CA, USA
*Researcher, Co-Founder & CTO* — *Mar 2024 – Jan 2025*
- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 3 research papers: quantum machine learning for EEG signal processing (2) and EEG-to-image diffusion (1).

**Department of Computer Science, National Yang Ming Chiao Tung University** — Hsinchu, Taiwan
*Research Assistant, Advisor: Prof. Chun-Shu Wei* — *Dec 2023 – Aug 2024*
- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (32 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

**MediaTek Inc.** — Hsinchu, Taiwan
*Digital IC R&D Engineer* — *Dec 2021 – Oct 2023*
- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implement.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

**Department of Surgery, National Taiwan University Hospital** — Taipei, Taiwan
*Research Assistant, Advisor: Dr. Shuo-Lun Lai* — *Jul 2021 – Sep 2021*
- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

**Department of Psychiatry, Taipei Veterans General Hospital** — Taipei, Taiwan
*Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li* — *Sep 2019 – Jun 2021*
- Built ML models on clinical EEG from major-depressive-disorder patients to predict treatment response.

**Max Planck Institute for Chemical Physics of Solids (MPI CPfS)** — Dresden, Germany
*Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao* — *Jul 2018 – Sep 2018*
- Synthesis and characterization of candidate Co-based chalcogenide superconductors.

**Other:** OpenAI (contractor via Mercor) — AI Trainer, RLHF preference data & model-behavior evaluation — *Mar 2025 – Oct 2025*

## Agent & LLM Engineering

**Berkeley RDI AgentX-AgentBeats — 3rd Place / 1,300+ teams (2026).** *MadGAA*, a multi-agent OSCE medical evaluation system: role-specialized agents (examiner, patient, judge) + verifiable scoring rubric (RAG + tool use + verifier). [code] [leaderboard]
**US-stock LLM agent suite (deployed).** Browser agent as an explicit PLAN→LOCATE→ACT→VERIFY state machine with fault-injection recovery; layered SEC 10-K extractor; 22 research agents + 11 placebo controls with lookahead-free backtests, per-provider cost ledger, and eval dashboards.
**Robotics Workcell Optimizer (live).** NL → spec → robot selection → 2D layout → 3D preview via multi-LLM abstraction (Claude/OpenAI/Gemini, validation-repair loop) + OR-Tools CP-SAT; ISO 13855 safety as inviolable hard constraints. [code] [demo]
**Also:** *flyhypo* (connectome + PubMed + LLM → grounded, falsifiable neuron hypotheses) [code]; *paper-evidence* (verbatim-quote gate + cross-family judge to block LLM hallucination) [code]; two live NYCU RAG physics tutors (Next.js + Gemini + pgvector) [live].
**Stack:** LangChain/LangGraph, ReAct & function calling, multi-agent orchestration, vLLM, LoRA/QLoRA, SFT/DPO/RLHF, RAG (pgvector/FAISS/Qdrant), OR-Tools, OpenAI/Anthropic/Gemini SDKs.

## Education

**National Taiwan University (NTU)** Taipei, Taiwan
*Master of Science in Biomedical Electronics and Bioinformatics* *Sep 2019 – Jun 2021*
*Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen*

- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.
- Master's thesis: "EEG Analysis for Prediction of Antidepressant Responses of Transcranial Magnetic Stimulation in Major Depressive Disorder Based on Attentional Convolution Time Series Prototypical Neural Network Model and Classical/Quantum Machine Learning Approaches."

**National Chiao Tung University (NCTU)** Hsinchu, Taiwan
*Bachelor of Engineering in Electrophysics* *Sep 2015 – Jun 2019*
*Bachelor of Science in Interdisciplinary Science Degree Program*

## Skills

**AI/ML:** Python, PyTorch, HuggingFace, Scikit-learn, MNE, MLflow, R, MATLAB
**Systems:** C/C++, JavaScript, Rust, MySQL, Solidity, Git, Docker, GCP, AWS
**Hardware:** Verilog, SystemVerilog, UVM, HSPICE, Genus, Verdi, SG-Lint, Perl
**Languages:** Chinese (Native), English (Fluent)

## Selected Publications

(49 total, 491 cites, h-index 13 — full list on Google Scholar)

[First author unless noted]

**Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting"** *Under review*, ICLR 2027; arXiv:2602.08768.

**Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model"** arXiv preprint (2025), arXiv:2502.17464 — **29 citations**.

**Chen, C.S., Wei, C.S. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning"** arXiv preprint (2024), arXiv:2406.16910 — **32 citations**.

**Chen, C.S., et al. "Exploring the Potential of Electroencephalography Signal–Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis"** JMIR Medical Informatics (2025), https://doi.org/10.2196/72027 .

**Chen, C.S., et al. "Exploring the potential of QEEGNet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning"** Journal of Signal Processing Systems 98, 5 (2026), https://doi.org/10.1007/s11265-025-01979-2 .

Li, C.T., **Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation"** Journal of Affective Disorders (2023), https://doi.org/10.1016/j.jad.2023.08.059 . [2nd author; model deployed at Taipei VGH PreDIC]

## Invited Talks & Presentations

**NVIDIA GTC 2026** (Top 8 Finalist, two posters); **MICCAI 2026** — DEMI spotlight talk and CLINICCAI poster; **IEEE ICASSP 2026** (two orals) and **2025** (oral); **IEEE Quantum Week 2025**; **DH2026** (two short papers); **QTML 2024** (poster).

**Guest lectures** at NYCU ECE (Wearable Chip Design), NTU CSIE (Advanced Compilers) and NYCU Maker Club; invited seminar, Dept. of Applied Physics, Tunghai University (2025–2026).

## Professional Service

**Conference Reviewer** (2026–2027): NeurIPS, ICML (Gold Reviewer — top tier), AAAI, KDD, MICCAI, ICASSP, NLDL 2027, ACM ICAIF 2026, IEEE QAI 2026. **Program Committee:** QNRL Workshop @ IEEE WCCI 2026.

**Journal Reviewer** (2025–present): 40 journals incl. TPAMI, TNNLS, TASLP, JBHI, Neural Networks, npj Quantum Information, Scientific Reports; **IOP Trusted Reviewer** (2026).

## Awards & Achievements

**2026 NVIDIA GTC Top 8 Finalist:** "GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB."

**2026 Berkeley RDI AgentX-AgentBeats:** Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

**2023 Certificate of the 20th National Innovation Award (Taiwan), clinical research category** — for the EEG treatment-response model now deployed at Taipei VGH PreDIC.

**2019 International Blockchain Olympiad (IBCOL):** World Finalist competition with 100+ teams.

**2017 Microsoft International Imagine Cup :** 2nd place in Taiwan.

**2016 International Genetically Engineered Machine Competition (iGEM):** World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.